

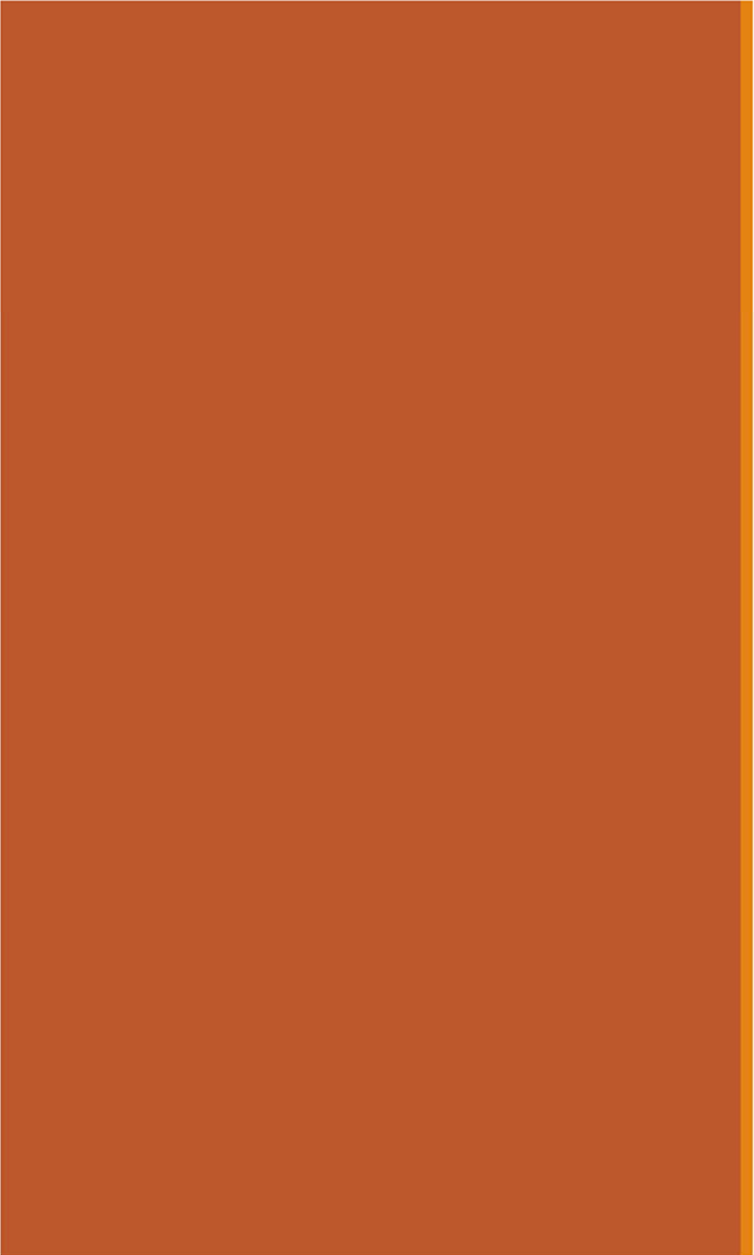
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15: General Inference

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February 12, 2024

[Lecture Discussion on Ed](#)

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

General Inference: Introduction

Inference

WebMD Symptom Checker WITH BODY MAP

INFO SYMPTOMS QUESTIONS CONDITIONS DETAILS TREATMENT

What are your symptoms?

Type your main symptom here

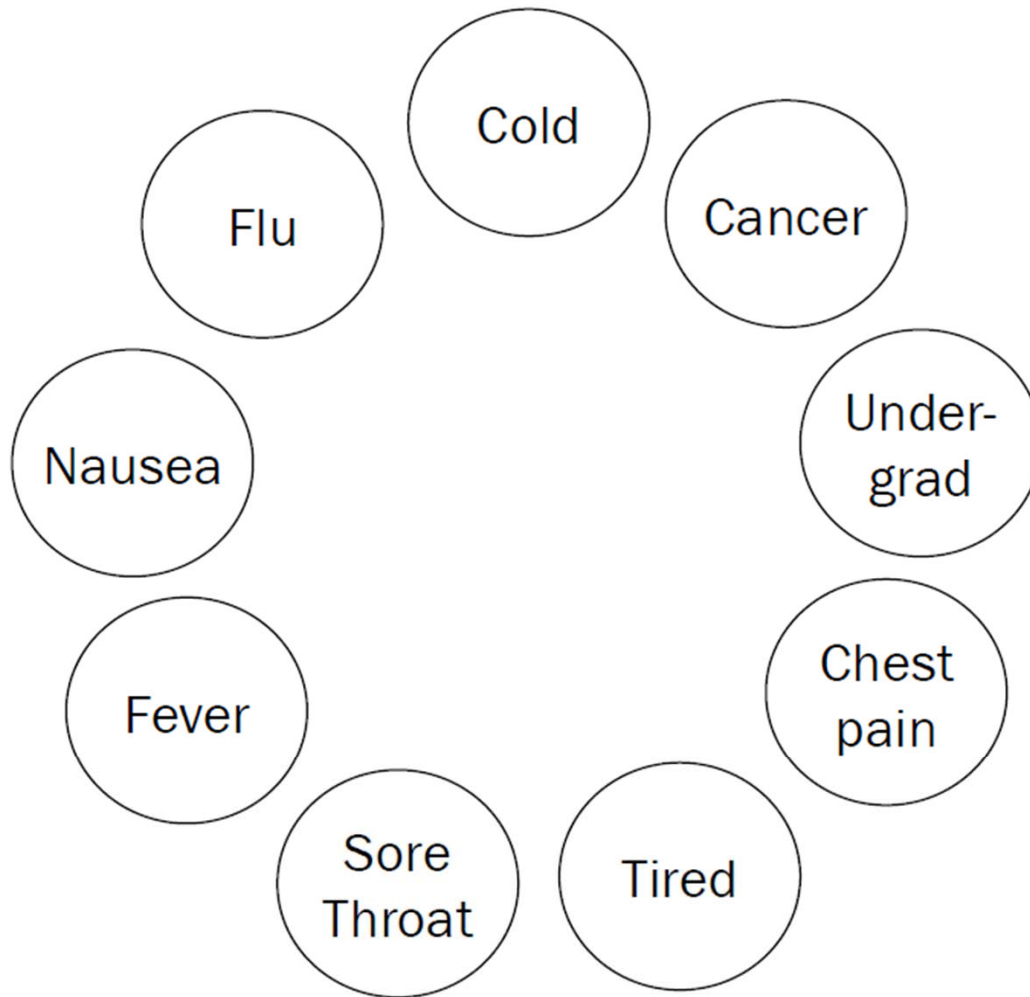
My Symptoms

- nausea
- fever 100.5f to 102f
- severe headache
- shaking chills

Results Strength: MODERATE



Inference

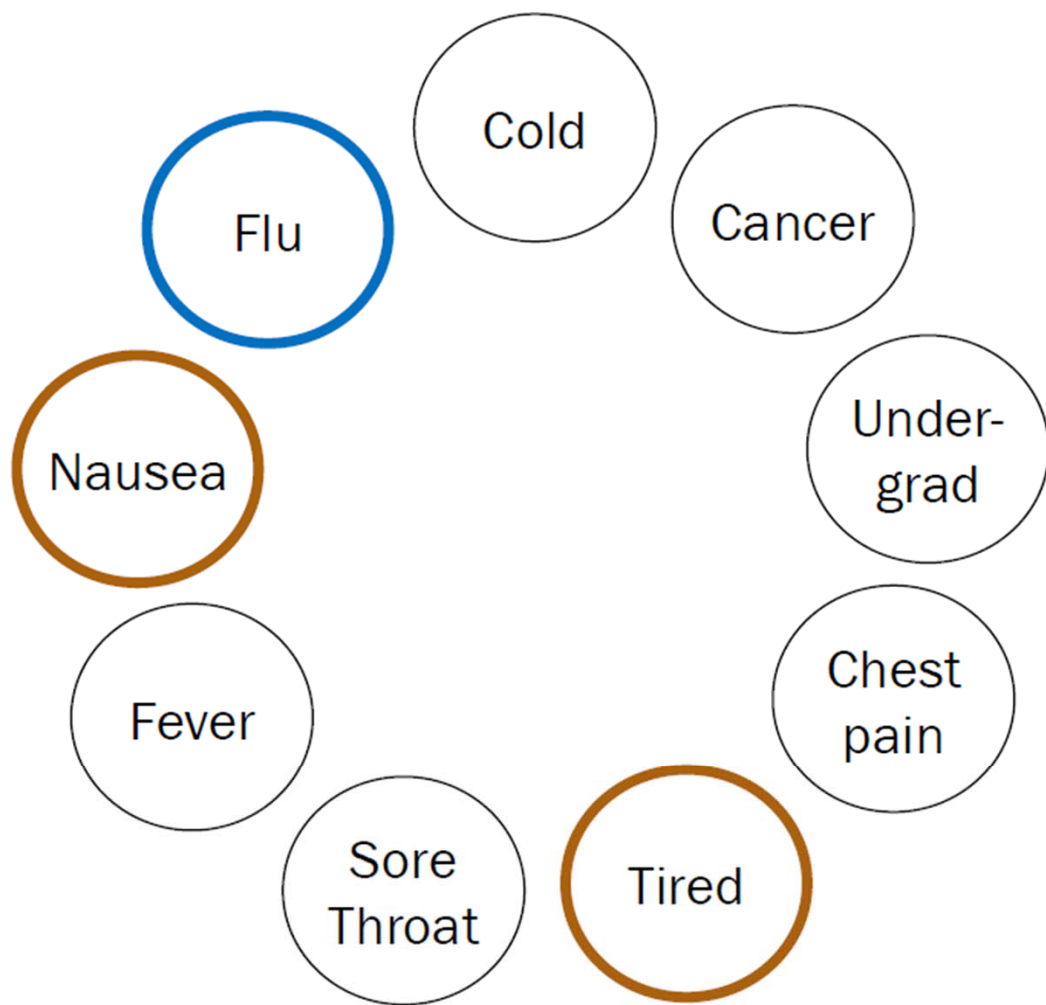


General inference question:



Given the values of some random variables, what is the conditional distribution of some other random variables?

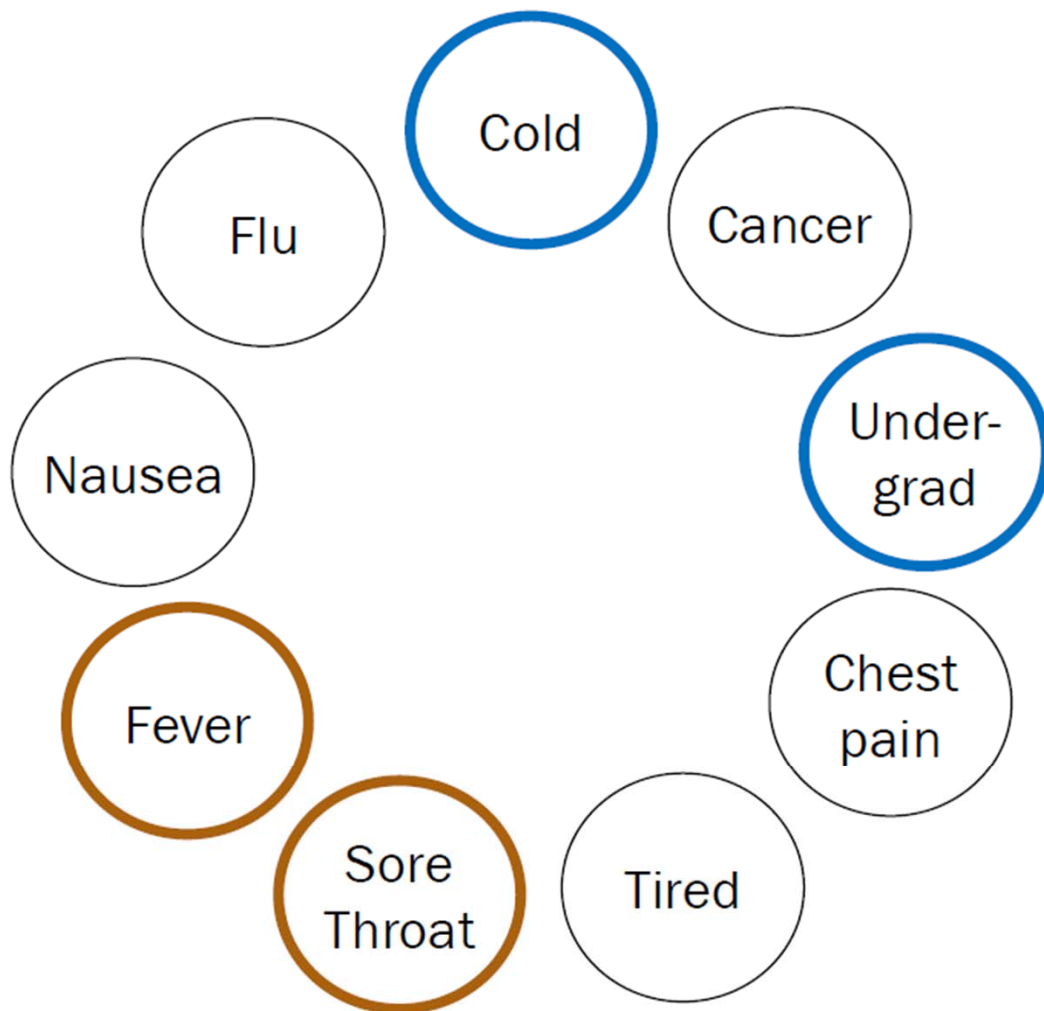
Inference



One inference question:

$$P(F = 1 | N = 1, T = 1)$$
$$= \frac{P(F = 1, N = 1, T = 1)}{P(N = 1, T = 1)}$$

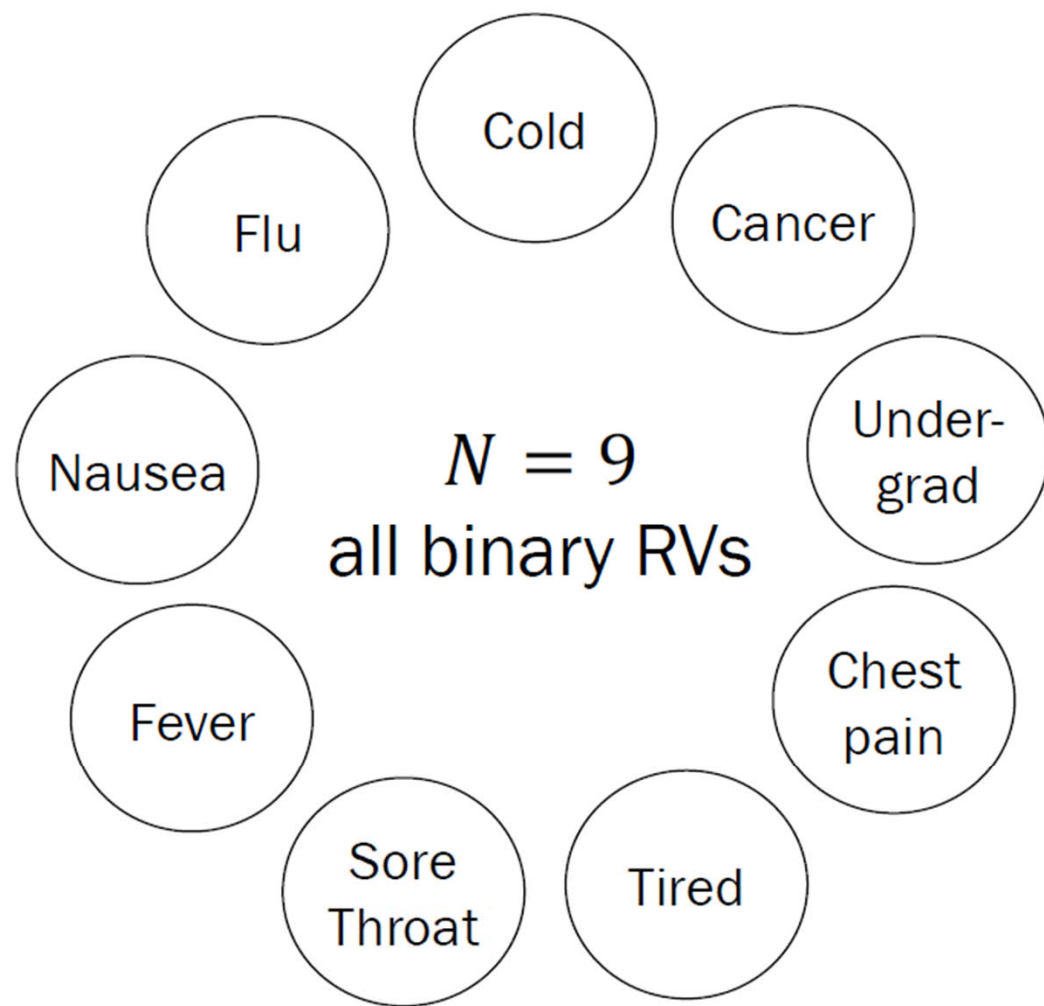
Inference



Another inference question:

$$P(C_o = 1, U = 1 | S = 0, F_e = 0) \\ = \frac{P(C_o = 1, U = 1, S = 0, F_e = 0)}{P(S = 0, F_e = 0)}$$

Inference



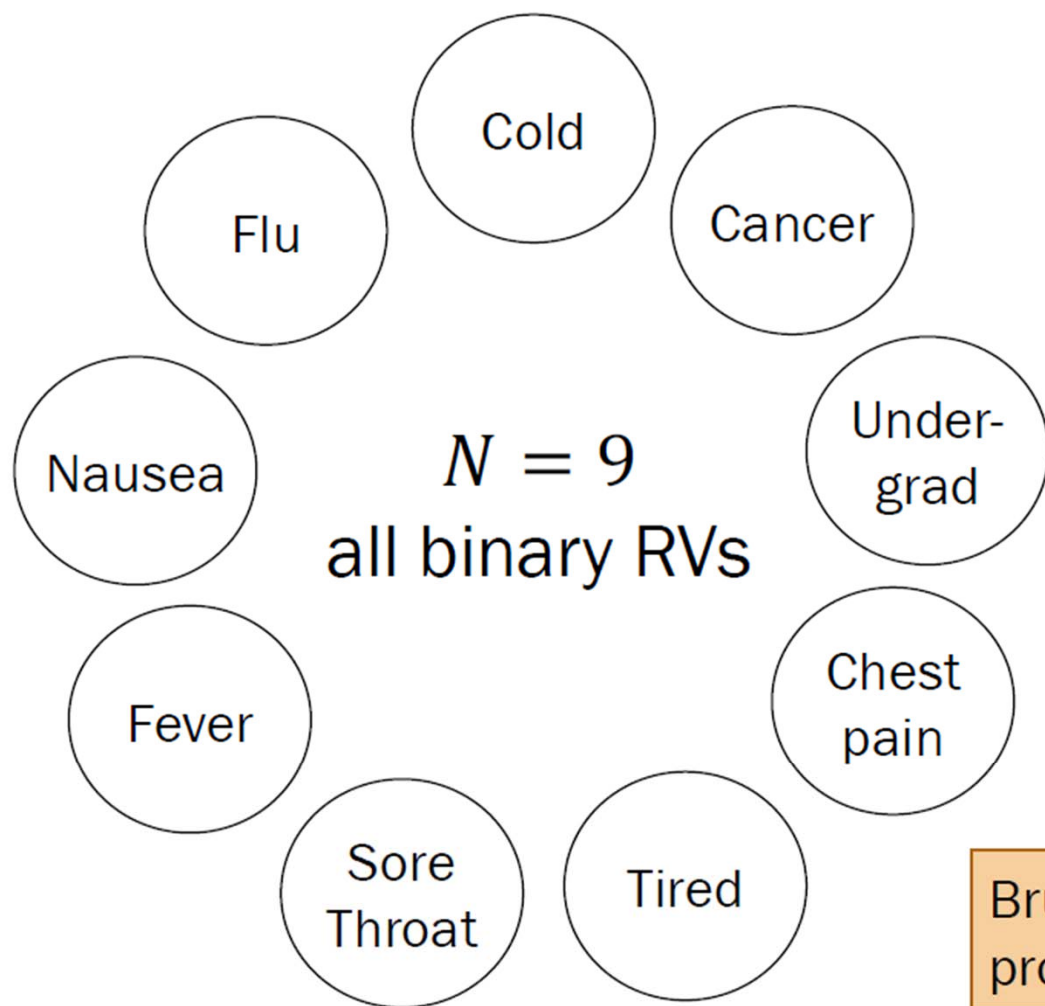
If we know the full joint distribution, we can answer all probabilistic inference questions.

What is the size of the joint probability table?

- A. 2^{N-1} entries
- B. N^N entries
- C. 2^N entries
- D. None/other/don't know



Inference



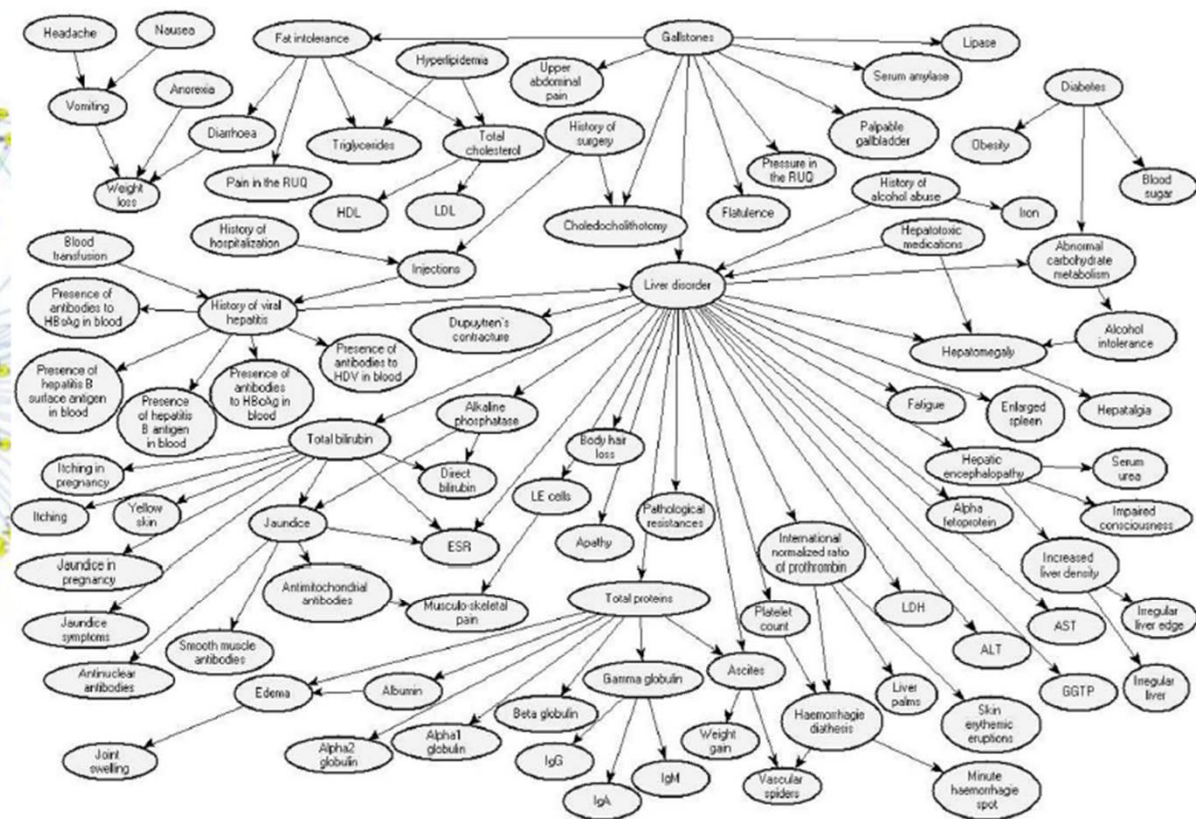
If we know the full joint distribution, we can answer all probabilistic inference questions.

What is the size of the joint probability table? why 2^N ?

- A. 2^{N-1} entries
 - B. N^N entries
 - ☒ C. 2^N entries
 - D. None/other/don't know
- Handwritten notes:*
P($x_1 = _, x_2 = _, \dots, x_N = _$)
= some value, but each underscore can be filled in in one of two ways.

Brute-force computation of a full joint probability mass function is often intractable. 다루기 힘들

A complex network graph visualization showing a large number of nodes (yellow circles) interconnected by edges. The edges are color-coded into three main groups: red, green, and blue. The red edges form a dense, highly interconnected cluster in the upper-left and central regions. The green edges form a large, dense cluster in the lower-left and central regions. The blue edges form a more sparse, elongated cluster on the right side of the graph. The overall structure is highly interconnected, with many nodes having multiple connections, suggesting a complex, possibly social or biological, network.



Conditionally Independent RVs

Recall that two events A and B are conditionally independent given E if:

$$\underline{P(AB|E)} = \underline{P(A|E)}\underline{P(B|E)}$$

n discrete random variables $X_1, X_2, \dots, X_n$ are called **conditionally independent given Y** if:

for all $x_1, x_2, \dots, x_n, y$:

when $n=2$ $P(X_1=x_1, X_2=x_2 | Y=y) = P(X_1=x_1 | Y=y) \cdot P(X_2=x_2 | Y=y)$

$$\star P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | Y = y) = \prod_{i=1}^n P(X_i = x_i | Y = y)$$

This implies the following (cool to remember for later):

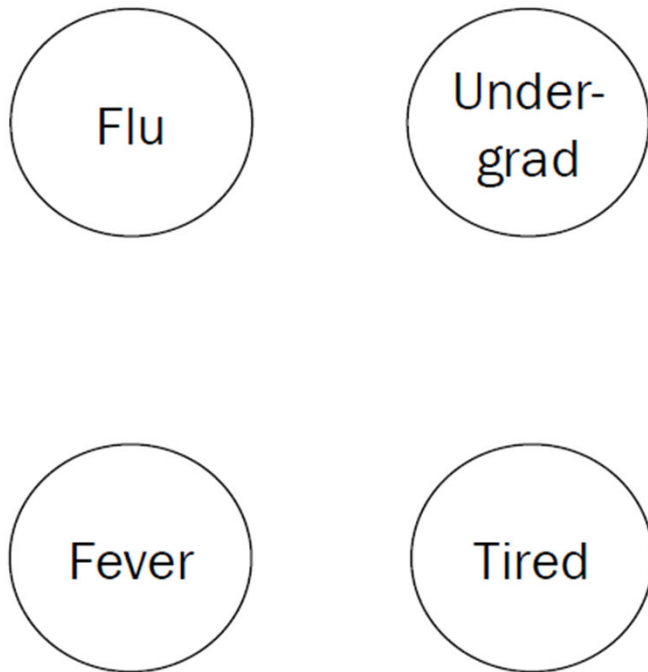
$$\log P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | Y = y) = \sum_{i=1}^n \log P(X_i = x_i | Y = y)$$

useful when N is large and probabilities are so small that numerical stability is a nck!



Bayesian Networks

A simpler WebMD



Great! Just specify $2^4 = 16$ joint probabilities?

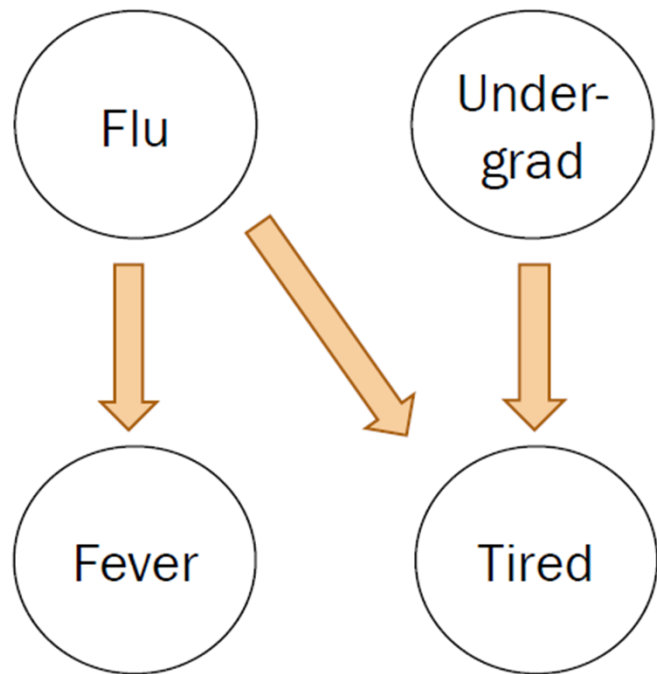
each take on either 0 or 1 independently of each other
 $P(F_{lu} = a, F_{ev} = b, U = c, T = d)$ $\rightarrow 2^4$ possibilities

What would an infectious diseases (ID) expert do?

인과관계

Describe the joint distribution using causality!

Constructing a Bayesian Network

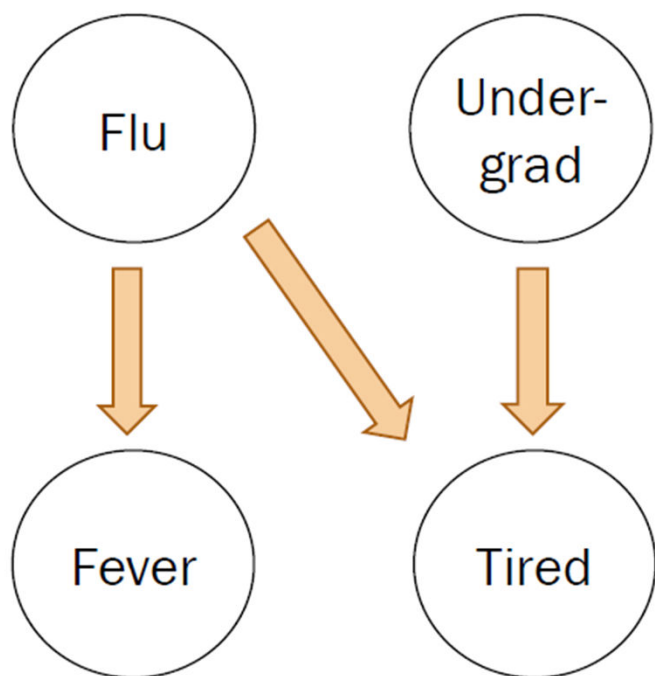


What would an ID expert do?

1. Describe the joint distribution using causality.

2. Assume
conditional
independence.

Constructing a Bayesian Network



In a Bayesian Network,
Each random variable is
conditionally independent of its
non-descendants, **given its parents**.

- fancy speak that says two variables are conditionally independent of each other if there's no path between them*
- Node: random variable
 - Directed edge: **conditional dependency**

in practice, a causal dependency as well

Examples: *doesn't inform Fev*

$$\textcircled{\bullet} P(F_{ev} = 1 | T = 0, F_{lu} = 1) = P(F_{ev} = 1 | F_{lu} = 1)$$

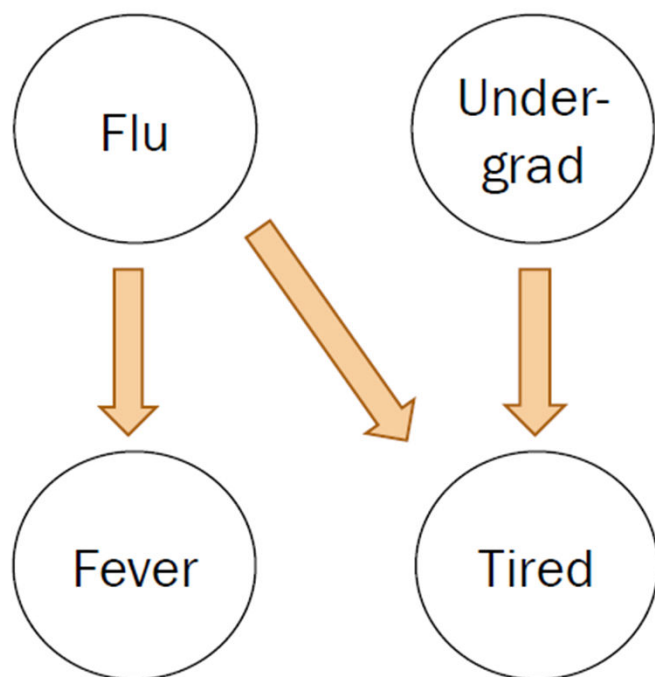
$$\textcircled{\bullet} P(F_{lu} = 1, U = 0) = P(F_{lu} = 1)P(U = 0)$$

no conditional dependencies expressed in the network, so independent

Constructing a Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



What would an ID expert do?

1. Describe the joint distribution using causality.
- ✓ 2. Assume conditional independence.
3. Provide $P(\text{values}|\text{parents})$ for each random variable

What conditional probabilities should our expert specify?

$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

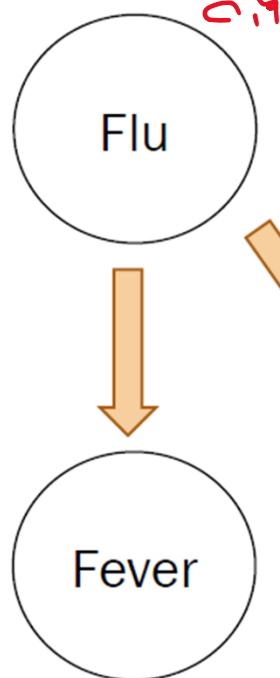
$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$



Constructing a Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$
$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

What would an ID expert do?

1. Describe the joint distribution using causality.
2. Assume conditional independence.
3. Provide $P(\text{values}|\text{parents})$ for each random variable

What conditional probabilities should our expert specify?

$$P(T = 1 | F_{lu} = 0, U = 0)$$

$$P(T = 1 | F_{lu} = 0, U = 1)$$

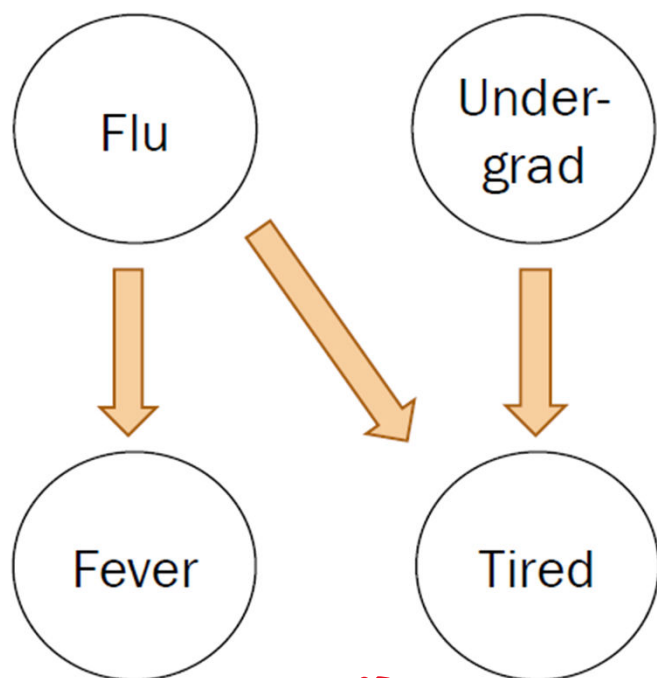
$$P(T = 1 | F_{lu} = 1, U = 0)$$

$$P(T = 1 | F_{lu} = 1, U = 1)$$

Using a Bayes Net

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

given

$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$

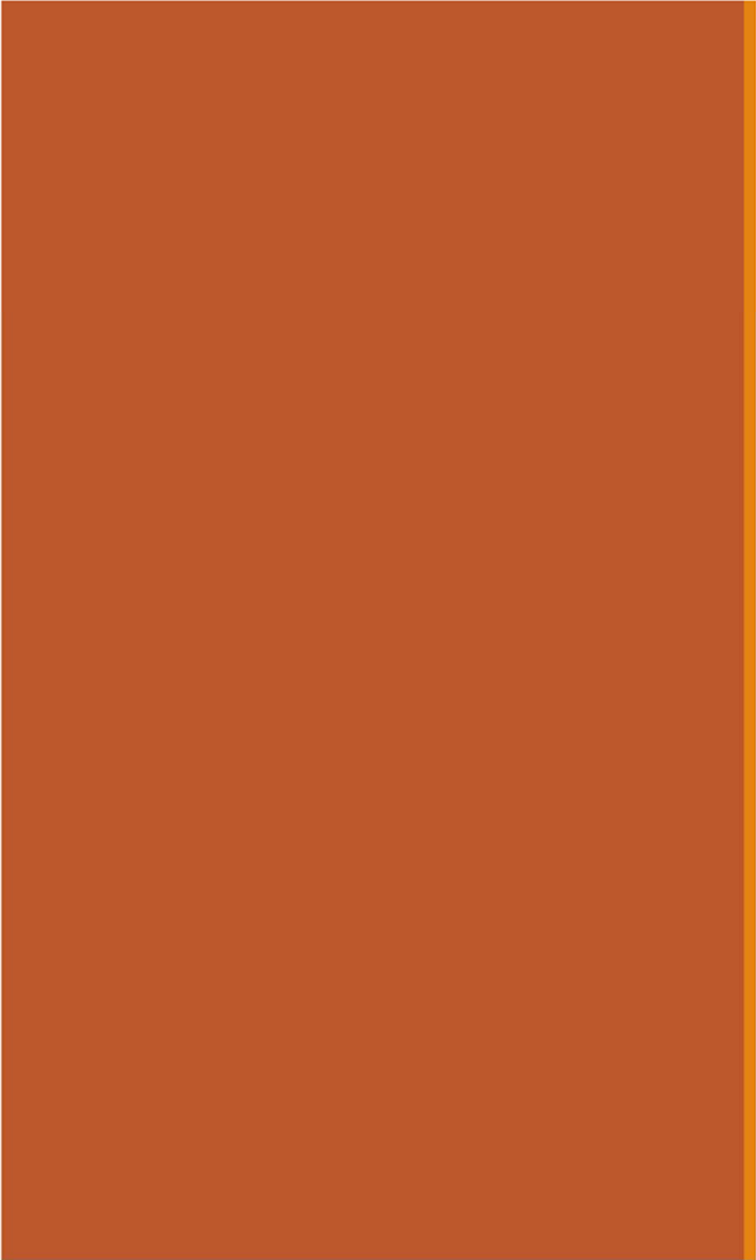
$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

What would a CS109 student do?

1. Populate a Bayesian network by asking an infectious diseases expert or by using reasonable assumptions

2. Answer inference questions

Our focus today



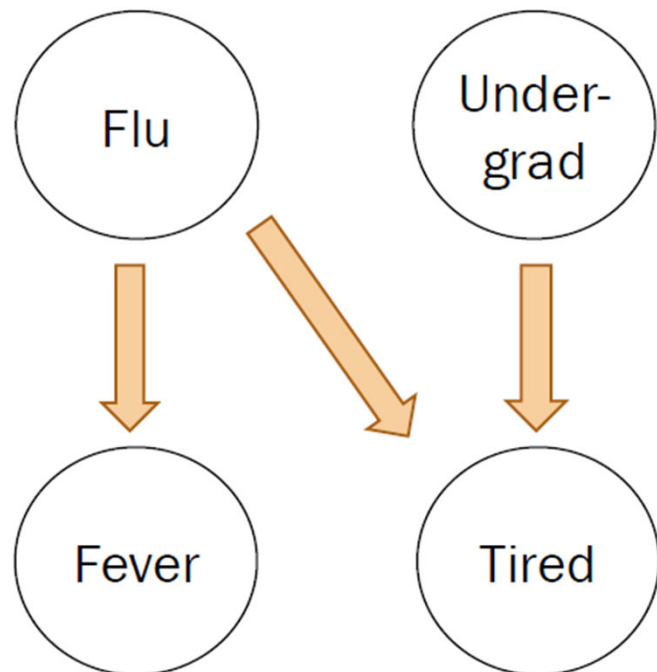
Inference: Math

Inference via math

Chap 4. Chain Rule (aka Product rule)
 $P(EF) = P(F) P(E|F)$

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

1. $P(F_{lu} = 0, U = 1, F_{ev} = 0, T = 1)$?

Compute joint probabilities
using chain rule.

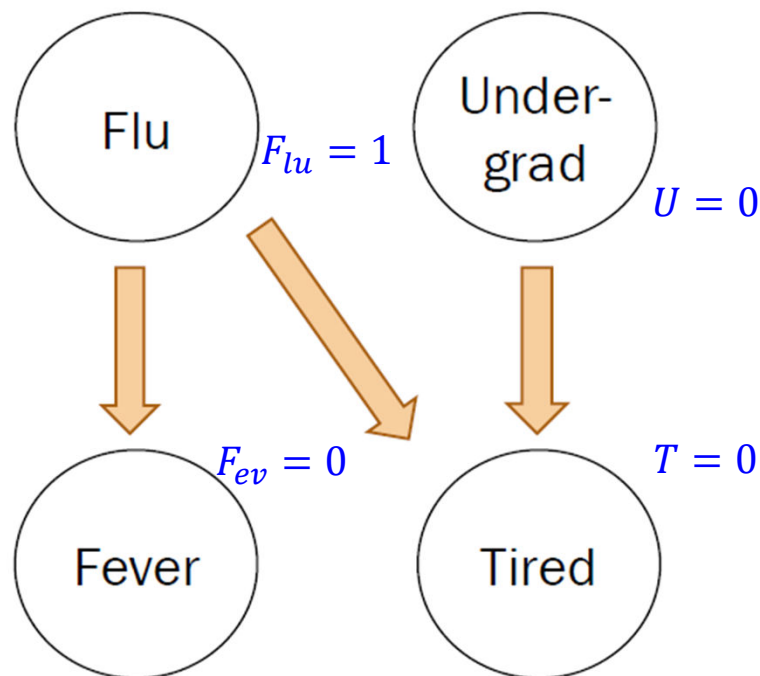
$$\begin{aligned}
 F_{lu} &\longrightarrow P(F_{lu} = 0) &&= 0.9 \\
 U &\longrightarrow P(U = 1) &&= 0.8 \\
 F_{ev} &\longrightarrow P(F_{ev} = 0 | F_{lu} = 0) &&= 0.95 \\
 T &\longrightarrow P(T = 1 | F_{lu} = 0, U = 1) &&= 0.8 \\
 &&&\underline{\underline{0.5472}}
 \end{aligned}$$

all four variables
contribute to joint probability value, but
how some contribute is influenced
by what they are
conditioned on.

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

2. ~~*~~ $P(F_{lu} = 1|F_{ev} = 0, U = 0, T = 1)?$

1. Compute joint probabilities

$$\checkmark P(F_{lu} = 1, F_{ev} = 0, U = 0, T = 1)$$

$$\checkmark P(F_{lu} = 0, F_{ev} = 0, U = 0, T = 1)$$

2. Definition of conditional probability

$$\frac{P(F_{lu} = 1, F_{ev} = 0, U = 0, T = 1)}{\sum_x P(F_{lu} = x, F_{ev} = 0, U = 0, T = 1)}$$

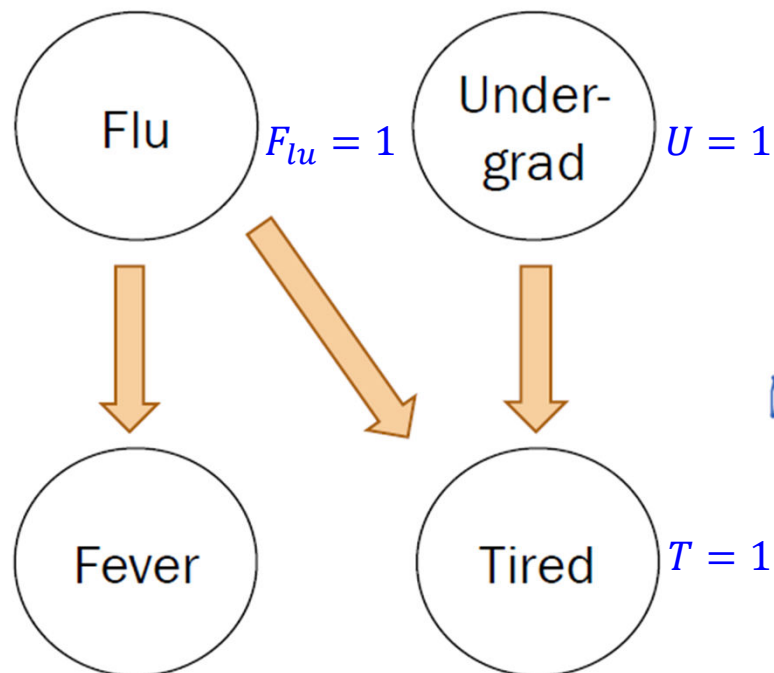
denominator results from application of law of total probability with one unknown

$$= 0.095$$

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

3. $P(F_{lu} = 1 | U = 1, T = 1)$ *conditional probability that ignores Fev*

1. Compute joint probabilities

there are four such probabilities here

$$P(F_{lu} = 1, U = 1, F_{ev} = 1, T = 1)$$

$$\dots$$

$$P(F_{lu} = 0, U = 1, F_{ev} = 0, T = 1)$$

2. Definition of conditional probability

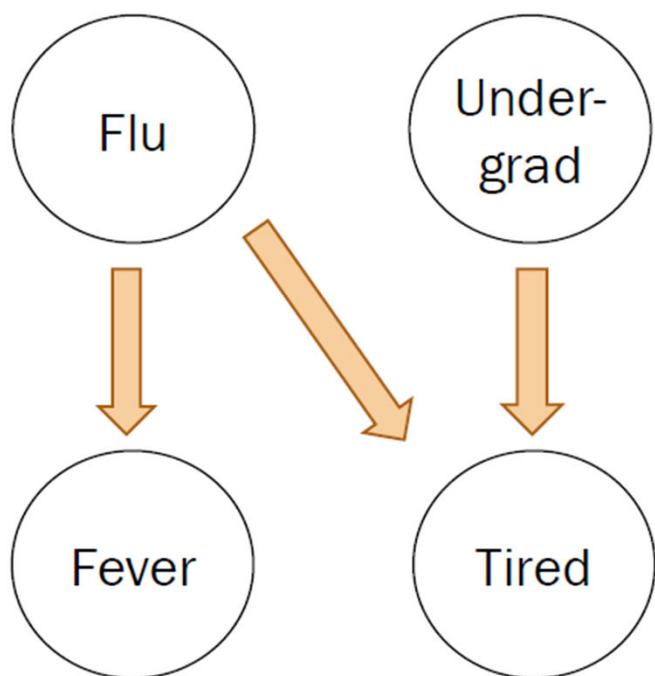
$$\frac{\sum_y P(F_{lu} = 1, U = 1, F_{ev} = y, T = 1)}{\sum_x \sum_y P(F_{lu} = x, U = 1, F_{ev} = y, T = 1)}$$

$$= 0.122$$

Inference via math

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



Solving inference questions precisely is possible, but sometimes tedious.

Can we use sampling to do approximate inference?

$$\begin{aligned} P(F_{ev} = 1|F_{lu} = 1) &= 0.9 \\ P(F_{ev} = 1|F_{lu} = 0) &= 0.05 \end{aligned}$$

$$\begin{aligned} P(T = 1|F_{lu} = 0, U = 0) &= 0.1 \\ P(T = 1|F_{lu} = 0, U = 1) &= 0.8 \\ P(T = 1|F_{lu} = 1, U = 0) &= 0.9 \\ P(T = 1|F_{lu} = 1, U = 1) &= 1.0 \end{aligned}$$



Rejection Sampling

정확한 확률밀도 함수를 알기 힘들거나 확률밀도 함수가 주어졌을 때, 해당 함수로부터 sample을 추출하기 어려운 경우 sampling 하는 기본적인 방법

[Rejection sampling - Wikipedia](https://en.wikipedia.org/wiki/Rejection_sampling)

https://en.wikipedia.org/wiki/Rejection_sampling

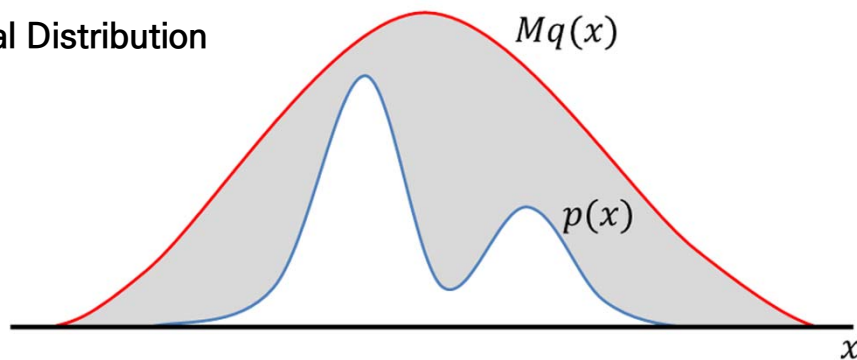
<https://untitledblog.tistory.com/134>

<https://untitledblog.tistory.com/134>

Rejection sampling의 기본적인 동작

- 쉽게 샘플을 생성할 수 있는 q 에서 샘플들을 생성한 뒤에 이 샘플들의 분포가 p 를 따르도록 수정하는 것
- 이를 통해 실제로는 q 에서 샘플이 생성되었지만, 그 결과는 p 에서 생성된 것처럼 만드는 것이다. 이때 쉽게 샘플을 생성할 수 있도록 임의로 설정한 q 를 제안 분포 (proposal distribution)이라고 한다.
- 제안 분포는 uniform distribution, normal distribution 등이 이용될 수 있으며, 가능하면 p 와 비슷한 형태의 확률 분포를 사용하는 것이 좋다.

1) Proposal Distribution



확률 분포 p 와 q, M 의 관계

제안 분포를 설정한 다음에는 상수 M 을 모든 x 에 대해 $p(x) \leq Mq(x)$ 이 되도록 설정한다.

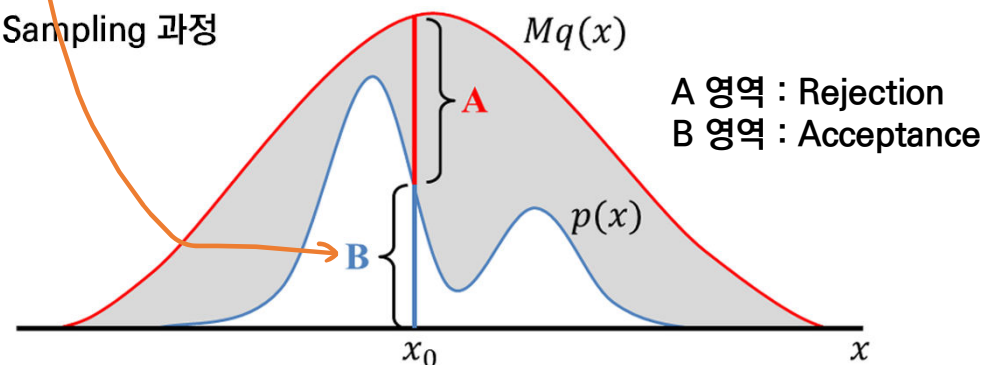
Algorithm 1: Rejection sampling

Input : the number of samples N ,
target distribution p ,
proposal distribution q ,
a given constant M

Output: samples $X = \{x_1, x_2, \dots, x_N\}$

```
1  $X = \{\}$ 
2 while  $n < N$  do
3    $x_0 \sim q(x)$ 
4    $u \sim U(0, 1)$ 
5   if  $u < \frac{p(x_0)}{Mq(x_0)}$  then
6      $X \leftarrow X \cup \{x_0\}$ 
7      $n \leftarrow n + 1$ 
8   end
9 end
```

2) Sampling 과정



확률 분포 q 로 부터 샘플 생성 및 Rejection / Acceptance 반복 수용

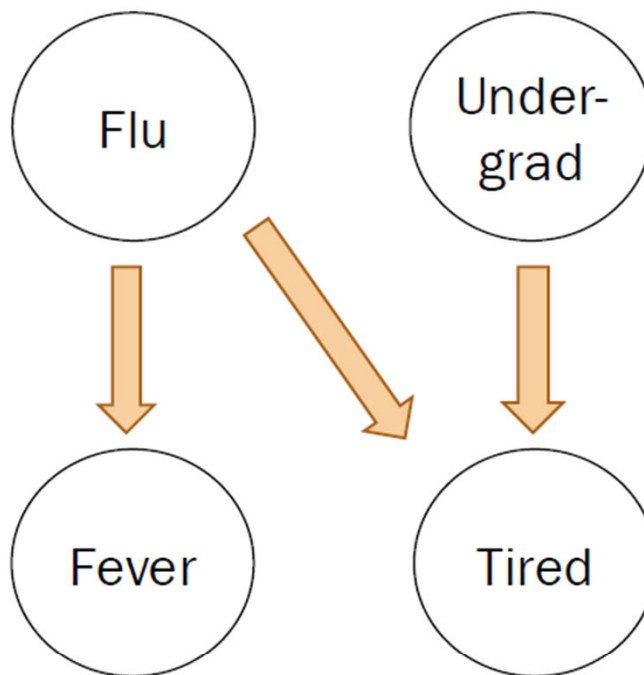
Rejection sampling algorithm

Step 0:

Require a fully specified
Bayesian Network

$$P(F_{lu} = 1) = 0.1$$

$$P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

Rejection sampling algorithm

this is the observation, or the given information

Inference question:

What is $P(F_{lu} = 1 | U = 1, T = 1)$?

[flu, und, fev, tir]

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
    # number of samples with (U = 1, T = 1)  
    samples_event = ...  
    # number of samples with (Flu = 1, U = 1, T = 1)  
    return len(samples_event) / len(samples_observation)
```

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    ✓ samples_observation = ...  
        # number of samples with  $(U = 1, T = 1)$   
    ✓ samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    ✓ return len(samples_event) / len(samples_observation)
```

$$\text{Probability} \approx \frac{\# \text{ samples with } (F_{lu} = 1, U = 1, T = 1)}{\# \text{ samples with } (U = 1, T = 1)}$$

Rejection sampling algorithm

Inference
question:

What is $P(F_{lu} = 1 | U = 1, T = 1)$?

$$\text{Probability} \approx \frac{\# \text{ samples with } (F_{lu} = 1, U = 1, T = 1)}{\# \text{ samples with } (U = 1, T = 1)}$$

Why would this definition of approximate probability make sense?

this is the frequentist's definition of probability!!

Why would this approximate probability make sense?

Inference
question:

What is $P(F_{lu} = 1 | U = 1, T = 1)$?

$$= \frac{P(F_{lu}=1, U=1, T=1)}{P(U=1, T=1)}$$

$$= \frac{\frac{n(F_{lu}=1, U=1, T=1)}{\text{num samples}}}{\frac{n(U=1, T=1)}{\text{num samples}}}$$

Why would this definition of approximate probability make sense?

$$\text{Probability} \approx \frac{\# \text{ samples with } (F_{lu} = 1, U = 1, T = 1)}{\# \text{ samples with } (U = 1, T = 1)}$$

Recall our definition of
probability as a frequency:

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

n = # of total trials
 $n(E)$ = # trials where E occurs

Rejection sampling algorithm

(1) samples=sample_a_ton()

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
    # number of samples with (U = 1, T = 1)  
    samples_event = ...  
    # number of samples with (Flu = 1, U = 1, T = 1)  
    return len(samples_event) / len(samples_observation)
```

[flu, und, fev, tir]

```
Sampling...  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 1, 0, 1]  
[0, 0, 0, 0]  
[0, 1, 0, 1]  
[0, 1, 1, 1]  
[0, 1, 0, 0]  
[1, 1, 1, 1]  
[0, 0, 1, 1]  
...  
[0, 1, 0, 1]  
Finished sampling
```

Rejection sampling algorithm

(1) samples=sample_a_ton()

```
N_SAMPLES = 100000
# Method: Sample a ton
# -----
# create N_SAMPLES with likelihood proportional
# to the joint distribution
def sample_a_ton():
    samples = []
    for i in range(N_SAMPLES):
        sample = make_sample() # a particle
        samples.append(sample)
    return samples
```

How do we make a sample
 $(F_{lu} = a, U = b, F_{ev} = c, T = d)$
according to the
joint probability?

Create a sample using the Bayesian Network!

Rejection sampling algorithm

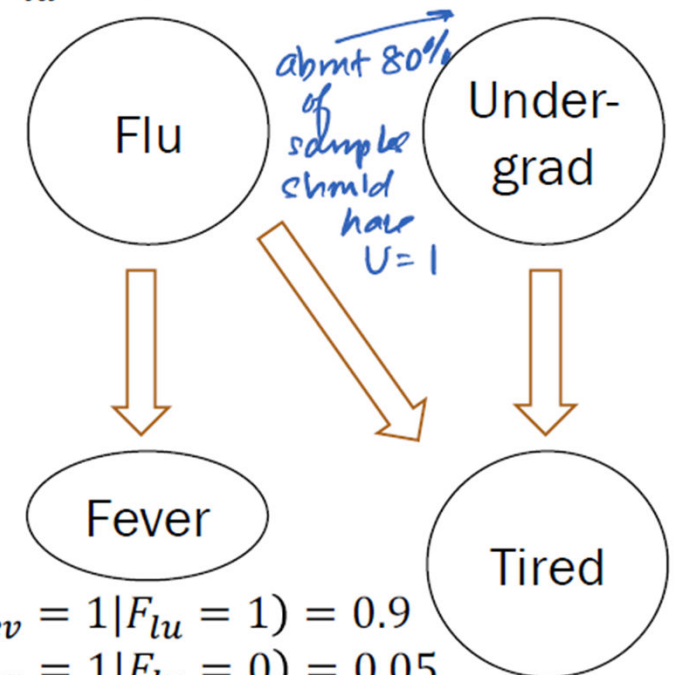
```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```

this suggests that about
10% of samples should
have $F_{lu} = 1$

$$P(F_{lu} = 1) = 0.1 \quad P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$

$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$

$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$

$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$

$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

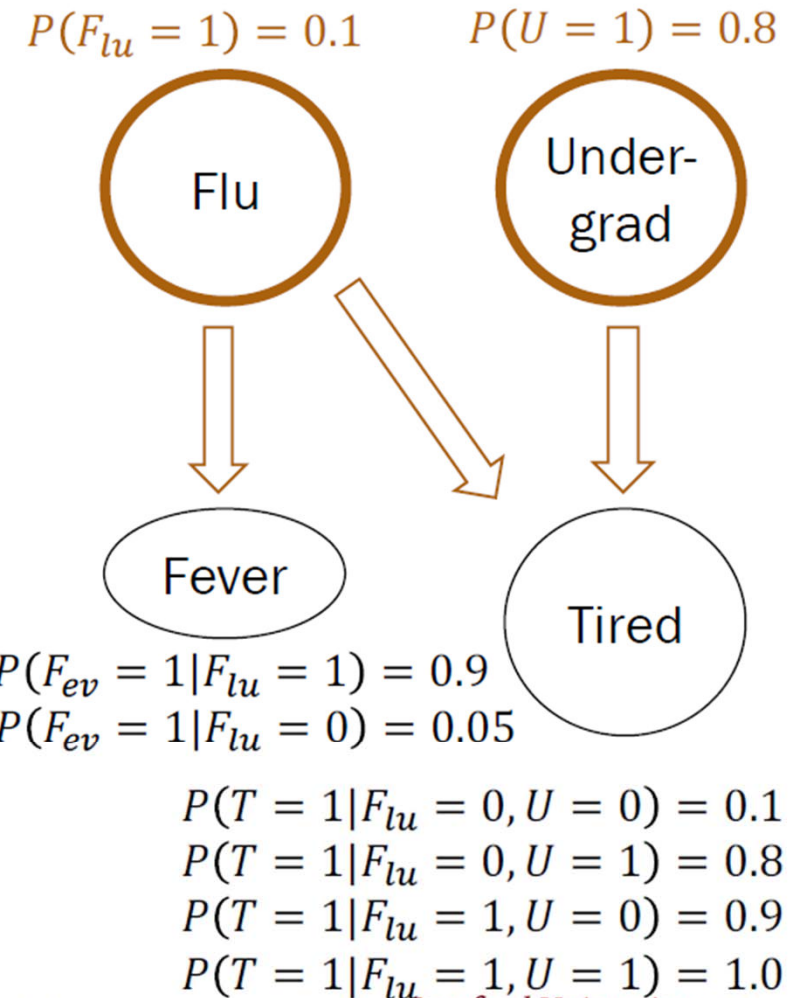
Rejection sampling algorithm

(1) samples=sample_a_ton()

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```



Rejection sampling algorithm

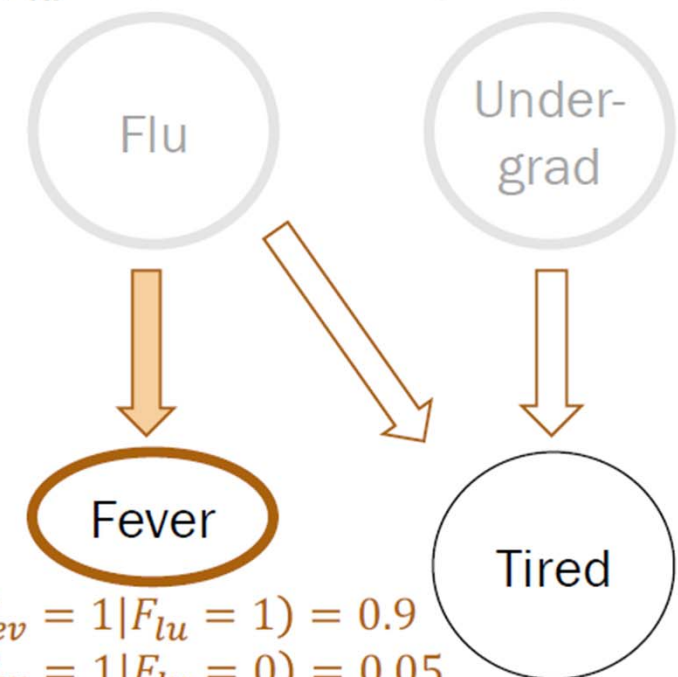
(1) samples=sample_a_ton()

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```

$$P(F_{lu} = 1) = 0.1 \quad P(U = 1) = 0.8$$



$$P(F_{ev} = 1 | F_{lu} = 1) = 0.9$$
$$P(F_{ev} = 1 | F_{lu} = 0) = 0.05$$

$$P(T = 1 | F_{lu} = 0, U = 0) = 0.1$$
$$P(T = 1 | F_{lu} = 0, U = 1) = 0.8$$
$$P(T = 1 | F_{lu} = 1, U = 0) = 0.9$$
$$P(T = 1 | F_{lu} = 1, U = 1) = 1.0$$

Rejection sampling algorithm

(1) samples=sample_a_ton()

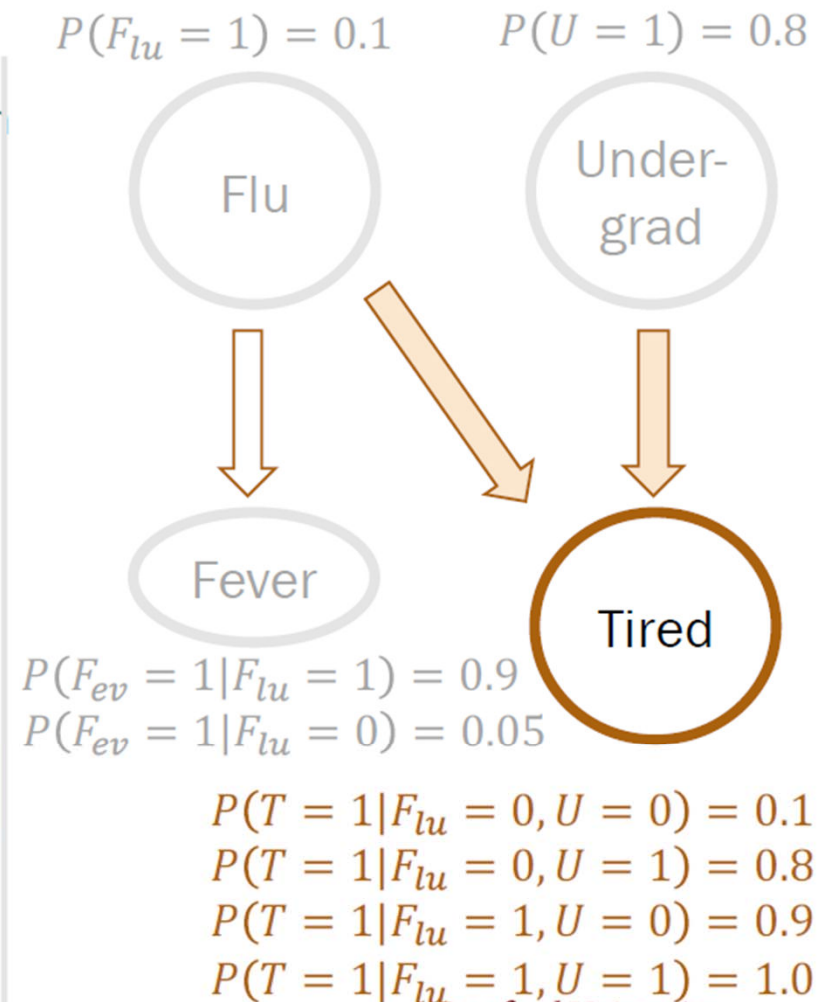
```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    #
    # TODO: fill in
    #
    #
    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tired]
```



Li Si Yang, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Winter 2023



Rejection sampling algorithm

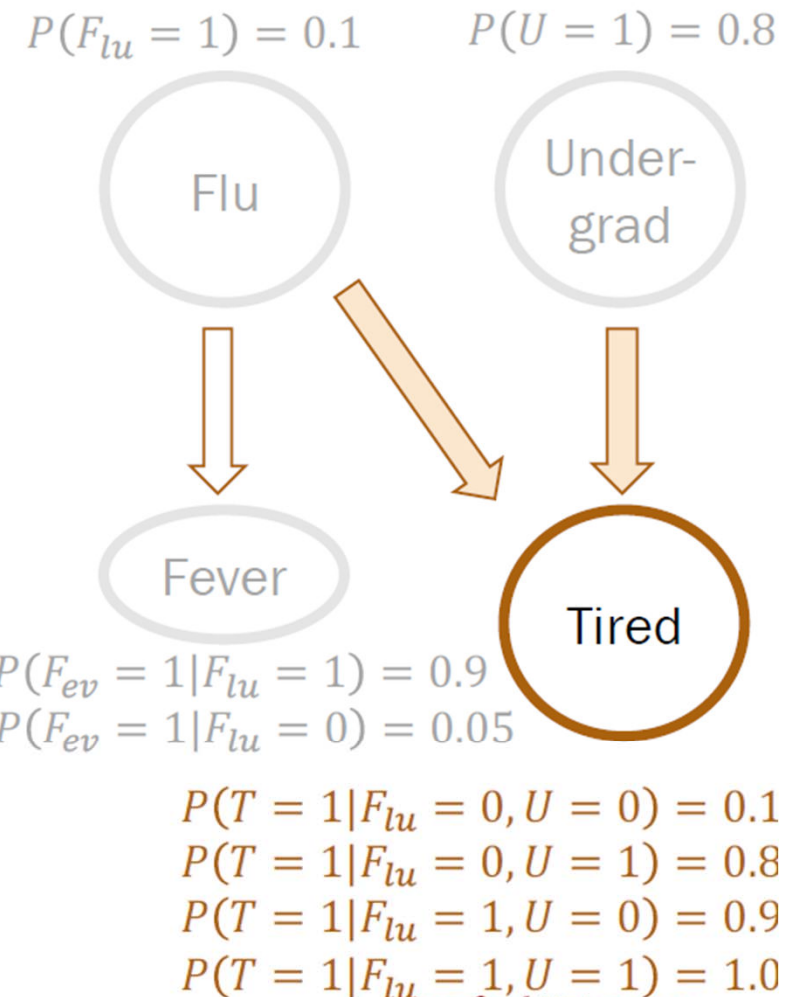
(1) samples=sample_a_ton()

```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    if flu == 0 and und == 0: tir = bernoulli(0.1)
    elif flu == 0 and und == 1: tir = bernoulli(0.8)
    elif flu == 1 and und == 0: tir = bernoulli(0.9)
    else: tir = bernoulli(1.0)

    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tir]
```



Rejection sampling algorithm

(1) samples=sample_a_ton()

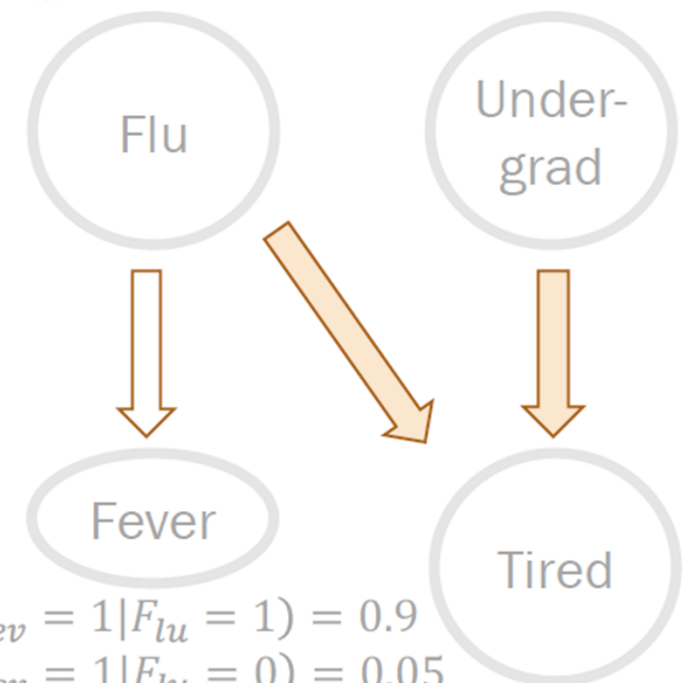
```
# Method: Make Sample
# -----
# create a single sample from the joint distribution
# based on the medical "WebMD" Bayesian Network
def make_sample():
    # prior on causal factors
    flu = bernoulli(0.1)
    und = bernoulli(0.8)

    # choose fever based on flu
    if flu == 1: fev = bernoulli(0.9)
    else: fev = bernoulli(0.05)

    # choose tired based on (undergrad and flu)
    if flu == 0 and und == 0: tir = bernoulli(0.1)
    elif flu == 0 and und == 1: tir = bernoulli(0.8)
    elif flu == 1 and und == 0: tir = bernoulli(0.9)
    else: tir = bernoulli(1.0)

    # a sample from the joint has an
    # assignment to *all* random variables
    return [flu, und, fev, tir]
```

$$P(F_{lu} = 1) = 0.1 \quad P(U = 1) = 0.8$$



$$P(F_{ev} = 1|F_{lu} = 1) = 0.9$$
$$P(F_{ev} = 1|F_{lu} = 0) = 0.05$$

$$P(T = 1|F_{lu} = 0, U = 0) = 0.1$$
$$P(T = 1|F_{lu} = 0, U = 1) = 0.8$$
$$P(T = 1|F_{lu} = 1, U = 0) = 0.9$$
$$P(T = 1|F_{lu} = 1, U = 1) = 1.0$$

Rejection sampling algorithm

(2) samples_observation

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = ...  
        # number of samples with  $(U = 1, T = 1)$   
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation = reject_inconsistent(samples, observation)  
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

build a new set of samples that retains just those samples consistent with the supplied observation.

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        # number of samples with  $(F_{lu} = 1, U = 1, T = 1)$   
    return len(samples_event) / len(samples_observation)
```

Keep only samples that are consistent
with the observation $(U = 1, T = 1)$.

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):
    samples = sample_a_ton()
    samples_observation =
        reject_inconsistent(samples, observation)
    samples
    # Method: reject_inconsistent
    # -----
    # Rejects all samples that do not align with the outcome.
    # Returns a list of consistent samples.
    return
    def reject_inconsistent(samples, outcome):
        consistent_samples = []
        for sample in samples:
            if check_consistent(sample, outcome):
                consistent_samples.append(sample)
        return consistent_samples
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        reject_inconsistent(samples_observation, event)  
    return len(samples_event) / len(samples_observation)
```

all these have $U=1, T=1$

further require this event to hold as well

Conditional event = samples with $(F_{lu} = 1, U = 1, T = 1)$.

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        reject_inconsistent(samples_observation, event)  
    return samples_event
```

Condition: $(F_{lu} = x, U = 1, F_{ev} = y, T = 1) \mid (F_{lu} = 1) = 1$.

```
def reject_inconsistent(samples, outcome):  
    consistent_samples =  
        [sample for sample in samples  
         if sample[F_{lu}] == outcome[F_{lu}]]  
    return consistent_samples
```

Rejection sampling algorithm

Inference question: What is $P(F_{lu} = 1 | U = 1, T = 1)$?

```
def rejection_sampling(event, observation):  
    samples = sample_a_ton()  
    samples_observation =  
        reject_inconsistent(samples, observation)  
    samples_event =  
        reject_inconsistent(samples_observation, event)  
    return len(samples_event)/len(samples_observation)
```

$$\text{Probability} \approx \frac{\# \text{ samples with } (F_{lu} = 1, U = 1, T = 1)}{\# \text{ samples with } (U = 1, T = 1)}$$

Rejection sampling

If you can sample enough from the joint distribution, you can answer any probability inference question.

With enough samples, you can correctly compute:

- Probability estimates
 - Conditional probability estimates
 - Expectation estimates
- but approximately, even if the approximation is incredibly good*

Why? Because your samples represent the joint distribution incredibly well!

[flu, und, fev, tir]

Sampling...

[0, 1, 0, 1]

[0, 1, 0, 1]

[0, 1, 0, 1]

[0, 0, 0, 0]

[0, 1, 0, 1]

[0, 1, 1, 1]

[0, 1, 0, 0]

[1, 1, 1, 1]

[0, 0, 1, 1]

...

[0, 1, 0, 1]

Finished sampling

$$P(\text{has flu} \mid \text{undergrad and is tired}) = 0.122$$